B Geeta Madhuri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.48
Resonance	Class XII (Telangana State Board of Intermediate Education)	2024	93.00%
Delhi School of Excellence	Class X (CBSE)	2022	95.00%

EXPERIENCE

Nickel Recovery & Porous Mesh Fabrication from Zn-Air Batteries | *IIT Hyderabad -- Ongoing*

- Developing a hydrometallurgical route for nickel recovery from spent Zn-air batteries using HSO/HO leaching.
- Electrodepositing recovered nickel onto polyurethane (PU) foam to fabricate a 3D porous mesh structure.
- Optimizing leaching and deposition parameters, including reaction time and current density, for improved recovery and coating quality.
- Studying the effect of deposition conditions on mesh morphology, pore structure, and mechanical integrity.
- Characterizing the recovered nickel mesh to assess deposition uniformity, porosity, and surface morphology.

PROJECTS

Biodiesel Production Process Simulation Using DWSIM | *(GitHub Link)*

DWSIM | *Process Simulation* | *Process Engineering* | *Thermodynamics*

Developed a steady-state DWSIM simulation of a continuous biodiesel production process by modeling the transesterification of triolein with methanol. Designed a process flowsheet incorporating feed preparation, mixing, preheating, and a conversion reactor, achieving **95% triolein conversion** to methyl oleate using the NRTL thermodynamic model. Configured process streams and operating conditions under steady-state assumptions and analyzed product-stream compositions to evaluate process performance. Verified simulation results through component-wise material balance checks to ensure physically consistent process behavior.

Heat Exchanger Design and Performance Analysis Tool Using Python | *(Link to Project)*

Python | *Heat Transfer* | *Matplotlib* | *DWSIM*

Developed a Python-based heat exchanger design tool for double-pipe heat exchangers using the LMTD and NTU-effectiveness methods. Implemented calculations for heat duty, heat transfer area, LMTD, capacity rates, capacity ratio, and effectiveness, with automated evaluation of the heat balance between hot and cold streams. Generated performance plots to analyze the effects of mass flow rate, overall heat transfer coefficient, and NTU on heat exchanger behavior. Validated the analytical results against DWSIM simulations by comparing heat duty, heat transfer area, LMTD, and effectiveness, demonstrating close agreement between the two approaches.

Distillation Column Design and McCabe-Thiele Analysis Tool | *(GitHub Link)*

Python | *Chemical Engineering* | *McCabe-Thiele Method* | *Process Design*

Developed a steady-state DWSIM simulation of a continuous biodiesel production process by modeling the transesterification of triolein with methanol. Designed a process flowsheet incorporating feed preparation, mixing, preheating, and a conversion reactor, achieving **95% triolein conversion** to methyl oleate using the NRTL thermodynamic model. Configured process streams and operating conditions under steady-state assumptions and analyzed product-stream compositions to evaluate process performance. Verified simulation results through component-wise material balance checks to ensure physically consistent process behavior.

TECHNICAL SKILLS

Programming: Python, C (Basic) | **Process Simulation:** DWSIM (Basic) | **CAD:** Solid Edge (Basic 3D) | **Libraries:**

Matplotlib, NumPy, Pandas

RELEVANT COURSES

- **Core Courses:** Introduction to Chemical Engineering, Applied Chemistry, Chemical Process Calculation, Transport Phenomena, Chemical Engineering Thermodynamics, Applied Mathematics in Chemical Engineering, Chemical Reaction Engineering - I, Heat Transfer, Fluid Mechanics.
- **Lab Courses:** Applied Chemistry Lab, Chemistry Laboratory, Physics Lab, Introduction to Programing.

POSITIONS OF RESPONSIBILITIES (PORS)

Internship Coordinator | Office of Career Services (FY 2026-27 | Ongoing)

- Coordinated with students, recruiters, and OCS team to support internship operations and facilitate recruitment.
- Reviewed student profiles and maintained placement records while enabling effective stakeholder communication.
- Streamlined internship-related workflows by supporting documentation, coordination, and timely resolution of queries.

Core Member | Otaku Club (2026-Present | Ongoing)

- Contributed to the planning and execution of various club events and workshops, ensuring smooth coordination.
- Actively participated in day-to-day club activities and supported the smooth operation of events across the academic year.
- Collaborated closely with fellow members to promote club activities and initiatives, helping boost overall participation.

Hospitality Coordinator | Milan - Inter-Hostel Sports & Cultural Fest (FY 2025-26)

- Coordinated logistics and on-ground operations for a large inter-hostel sports competitions and cultural festival.
- Managed participant coordination and event logistics, contributing to efficient operational delivery.
- Collaborated with multiple organizing teams to ensure seamless coordination and timely execution of activities.

Corporate Partnerships & Growth Core Member | Finance & Consulting Club (FY 2024-25)

- Contributed to the execution of Horizon through operational coordination and support across event activities.
- Participated in case discussions and collaborated on Corporate Partnerships & Growth initiatives.
- Assisted in execution activities for club initiatives while coordinating with team members for smooth operations.